

Problem Set MT1

Instructor: Daniele Micciancio

Due on: never

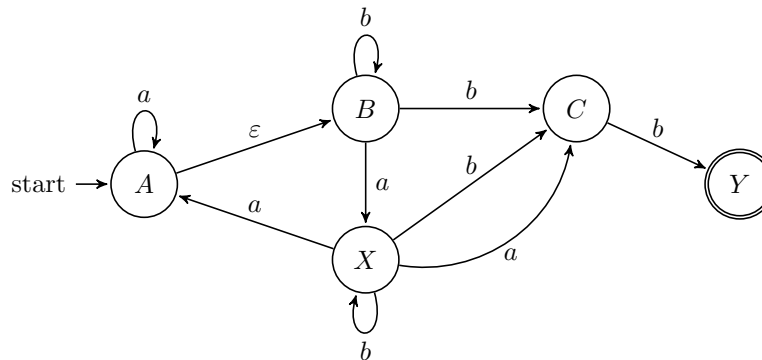
These are practice problems for Midterm 1, and cover the same material as HW1, HW2, HW3, plus context free grammars. (Push Down Automata will not be in the midterm exam.) You are not required to submit your answers. The aim is to cover all topics that may show up in the midterm exam. The actual midterm may be shorter and cover only a selection of topics.

Solutions to these practice problems will not be provided, but if you want to check your solutions, you are welcome to come to any of the additional office hours scheduled before the midterm. If you really want to read some problem solutions, you can do the following: feed the problems to an AI model, and then critically evaluate its answers. Just reading solutions to a given list of problems is not very useful, because the problems on the exams are very likely to be different. On the other hand, being able to evaluate the correctness of a solution can prove very valuable to check that your own work is correct.

Additional problems with solutions can be found at the end of each chapter in the textbook.

1 Tracing NFAs

Consider the following NFA:



For each of the following input strings, determine if the string is accepted by the automaton. If the string is accepted, give also an accepting computation. If it not accepted, give the set of all states the automaton can be in after reading the input.

1. ϵ
2. aa
3. bb
4. $aabb$.
5. $abab$.

2 Building DFAs

Provide a formal description and state diagram for a DFA over the alphabet $\Sigma = \{a, b\}$ that accepts the language $L = \{w \mid w \text{ contains an even number of } a\text{'s and at least one } b\}$.

3 NFA to DFA

Given an NFA with states $Q = \{q_0, q_1\}$, alphabet $\Sigma = \{0, 1\}$, start state q_0 , accept state q_1 , and transitions:

- $\delta(q_0, 0) = \{q_0, q_1\}$
- $\delta(q_0, 1) = \{q_1\}$
- $\delta(q_1, 0) = \emptyset$
- $\delta(q_1, 1) = \{q_0, q_1\}$

Construct the equivalent DFA using the subset construction algorithm. List the states of the DFA and its transition table.

4 NFA to Regex

Transform the NFA from the previous problem to a corresponding regular expression using the procedure studied in class.

5 NFA Design

Provide a formal description and state diagram for an NFA that accepts the language

$$L = \{0^n \mid n \text{ is a multiple of 5 or 7}\}.$$

Make use of non-determinism, so that your solution has a reasonably small number of states. How many states does your NFA use? Do you think this is the smallest size NFA for the language?

Do you think there is a DFA for the same language with a similar number of states? Provide an upper bound on the number of states required by an equivalent DFA, and justify your answer. What do you think is the smallest number of states required by a DFA for this language? (No proof required, just an informal justification.)

6 Regex Design

Write a regular expression for each of the following languages:

- All strings over $\Sigma = \{0, 1\}$ that do **not** contain the substring 110.
- All strings over $\Sigma = \{a, b, c\}$ that do not contain ac , ba or cb .

7 Regex Comparison

Consider the regular expressions $R = (a \cup b)^*$ and $S = (a^*b)^*$. Determine if $\mathcal{L}(R)$ is a subset, superset, or equal to $\mathcal{L}(S)$. Prove your answer.

8 CFG Design

Construct a Context-Free Grammar for the language $L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i + j = k\}$.

9 Grammars and Parsing

Consider the grammar

$$S \rightarrow ASA \mid aB$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b \mid \epsilon$$

with start symbol S and alphabet $\Sigma = \{a, b\}$, and answer the following questions:

- Can the empty string ϵ be generated by the grammar? Justify your answer.
- Give both a parse tree and a leftmost derivation for the string aa .
- Give a string of length 3 that is generated by the grammar, and justify your answer with a parse tree.
- Give a string of length 3 that cannot be generated by the grammar, and justify your answer.

10 Pumping Lemma

Use the Pumping Lemma for regular languages to prove that $L = \{ww^R \mid w \in \{0, 1\}^*\}$ is not regular (where w^R is the reverse of w).

11 Basic Closure Properties

We studied that regular languages are closed under the following operations: set union, set complement, concatenation, star, reverse.

Use these basic operations to prove that regular languages are closed under the following operation

$$T(L) = \{w \mid \text{either both } w \text{ and } w^R \text{ are in } L, \text{ or neither of them is in } L\}.$$

12 Closure Properties

Prove that regular languages are closed under the following operation:

$$T(L) = \{w \mid \text{all proper prefixes of } w \text{ belong to } L\}.$$

13 Challenge Problem

Consider the following operation:

$$\text{Half}(L) = \{x \mid \exists y \text{ such that } |x| = |y| \text{ and } xy \in L\}.$$

In other words, $\text{Half}(L)$ is the set of words obtained by taking a word $xy \in L$ of even length $2n$, and dropping the second half of the word. Prove or disprove that regular languages are closed under the Half operation. (As by the title, this is meant to be a harder problem. Good for training. Even if you do not manage to solve it, just trying may improve your understanding.)